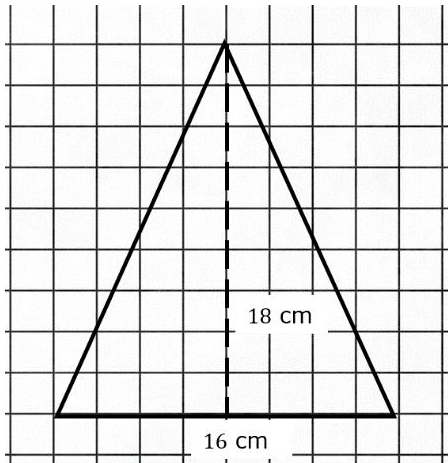


Grade 7
Unit 5
Lesson 9 Practice

Name Answer Key
Date _____
Period _____

A Pre-Algebra class created a scale drawing of the isosceles triangle shown below. The class will use the drawing to build an actual isosceles triangle for their Unit project. Each gridline on the map represents 2 centimeters (cm). Use this information to answer questions 1-4.



1. What is the area of the isosceles triangle on the scale drawing?

$$\frac{1}{2} \cdot 18 \cdot 16 = 144 \text{ cm}^2$$

2. If the scale on the map is $\frac{5 \text{ in}}{2 \text{ cm}}$, what is the area of the isosceles triangle created by the class?

$$144 \cdot \left(\frac{5}{2}\right)^2 = 900 \text{ in}^2$$

3. How does the area of the isosceles triangle built by the Pre-Algebra class compare to the area of the scale drawing? Explain your answer.

The triangle built by the class is 6.25 times larger which is $\left(\frac{5}{2}\right)^2$.

4. If the class want built another isosceles triangle with an area of 1296 in^2 , what scale factor was used based on the dimension of the scale drawing above?

$$\frac{1296}{144} = 9 \quad 9 = 3^2$$

The scale factor was 3.

5. Monique had an inground rectangular pool installed in her new home. She also purchased a safety cover for the pool. The actual area of the pool safety cover is $784 \text{ square feet (ft}^2\text{)}$. The area of the scale drawing for the safety cover is 16 ft^2 . What scale did Terrica use to create the scale drawing of her pool?

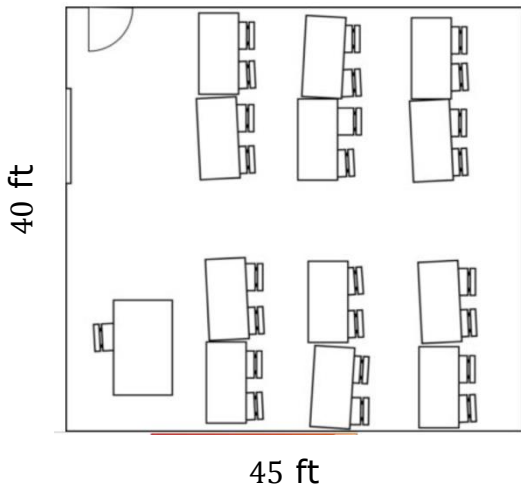
$$\frac{16}{784} = \frac{1}{49} \quad \frac{1}{49} = \left(\frac{1}{7}\right)^2 \quad \text{The scale is } \frac{1}{7} \text{ the size of the actual}$$

6. Monique wants to create a scale drawing of her entire property. The rectangular-shaped property is 250 feet (length) by 125 feet (width). If Monique uses a scale factor of $\frac{55 \text{ ft}}{1 \text{ in}}$, what area would her property be on the scale drawing?

$$250 \cdot 125 = 31250 \text{ ft}^2$$

$$\frac{31250}{55^2} = 10 \frac{40}{121} \text{ in}^2 \text{ or } \approx 10.3 \text{ in}^2$$

A 6th grade classroom is shown below. The actual dimensions of the classroom are 40 feet (length) by 45 feet (width). Use this figure to answer questions 7-10.



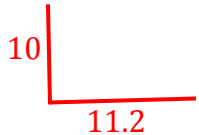
7. Brandon sketched a scale drawing of the classroom. The scaled drawing of the classroom has an area of 112.5 ft^2 . What scale factor did Brandon use to sketch the drawing?

$$40 \cdot 45 = 1800 \text{ ft}$$

$$\frac{112.5}{1800} = \frac{1}{16} \quad \frac{1}{16} = \left(\frac{1}{4}\right)^2$$

The scale is $\frac{1}{4}$ the size of the actual.

8. What are possible dimensions for the scale drawing (length and width)?



9. Brandon believes that if the area of the classroom in the figure was cut in fourths, the scale factor (found in #7) would equal be one fourth of the value. Is Brandon correct? Explain.

False, the scale factor needs to be squared.

10. The principal wants to build a band classroom double the size of the 6th grade classroom, what is the proposed area of the band classroom?

$$40 \cdot 45$$

↓

$$1800 \cdot 2^2 = 7,200 \text{ ft}^2$$